

From HCI Classroom to Complex Challenges: Enhancing HCI Education through Cross-Disciplinary Teamwork and Stakeholder Engagement

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Abstract

In this work, we present three pedagogical strategies designed to innovate HCI education using a challenge-driven approach. Rather than focusing solely on connecting HCI principles to single-user interaction problems, students 1) explore HCI problems within a cross-disciplinary challenge space, 2) work with diverse community stakeholders beyond end-users, and 3) learn to collaborate cross-functionally. Based on our case study of a semester-long HCI and design course that connects three design problems – spatial design, interactive systems design, and learning design – university students learned to apply design and evaluation processes across multiple disciplinary domains – including materials science, makerspace design, and learning sciences. They also recognized the complexity of HCI work through stakeholder engagement and collaborative strategies to work cross-functionally. We offer practical recommendations for implementing these pedagogical features in the HCI classroom to better prepare students for professional, complex real-world HCI practice.

CCS Concepts

• **Human-centered computing** → **Empirical studies in HCI**.

Keywords

HCI education, cross-disciplinary learning, stakeholder engagement, cross-functional teams, HCI professionals

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1 Introduction

Human-Computer Interaction (HCI) and Computer-Supported Cooperative Work (CSCW) pedagogy has traditionally focused on teaching HCI principles and design processes through one-off, isolated learning activities involving a highly focused HCI and design

problem. In the professional context, however, HCI professionals encounter complex problems where they need to understand HCI within a cross-disciplinary and sector-specific problem while equally seeking contributions as HCI and design professionals. Recent studies of HCI education have explored more experiential pedagogical approaches such as studio-based [17, 19, 28], and project-based [33, 34] learning to reflect this reality. Yet, studies need to examine more holistic approaches that better prepare students for the complex, multifaceted nature of HCI work in practice, nurturing students to make sense of technology or material use, interaction, and design beyond a confined scope.

This paper presents three pedagogical strategies implemented to engage students in exploring complex, real-world HCI and design problems: 1) *embedding HCI within a cross-disciplinary problem space*, 2) *engaging stakeholders beyond end-users*, and 3) *utilizing cross-functional team structure for collaboration*. Through a case study of a semester-long, challenge-driven HCI and design course in a research university, we highlight both the features of the implemented pedagogies and student learning experiences in spatial and educational design for underserved youth at a community tech center. We present our findings in two sections, highlighting three pedagogical strategies and associated student learning experiences (i.e., *grasping domain-specific knowledge in the broader challenge space*, *recognizing co-evolving dynamics of HCI and design through stakeholder engagements*, and *aligning processes and work across the functional teams*).

Our study makes two key contributions to the HCI community. First, it explores the role of HCI education and practice in the broader scholarly discussions on ‘convergence’ and ‘transdisciplinarity’. HCI is often situated within, and drives, increasingly complex, cross-domain problems, requiring the deep integration and a transdisciplinary approach to innovation [25, 37, 42]. The U.S. National Science Foundation, for instance, has prioritized facilitating transdisciplinary research and collaboration to better address scientific questions or societal needs [27]. This presents an opportunity for the HCI community to innovate educational approaches that can support converging problem areas. Second, our findings offer concrete recommendations for enhancing HCI teaching and learning, addressing the ongoing gap between HCI theory and practice [7, 38]. By incorporating cross-disciplinary, stakeholder-engaged approaches and leveraging professional cross-functional team structures, we show how these teaching strategies can bring provide students with different kinds of learning experiences that connect HCI to complex real-world applications.

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2 Background

2.1 HCI pedagogy for real-world complexity

Given that the interaction between people, technology, and infrastructure is central to HCI practice and research, a transdisciplinary approach to understanding this complexity has been increasingly used for framing HCI education [25, 42]. Yet, given the commonly used methods for practicing human-centered design and technology innovation, HCI pedagogy has traditionally focused on specific user experiences or technology design, often relying on one-off interactions involving particular users, tasks, and technology use.

HCI and design challenges become increasingly multifaceted, as HCI researchers and practitioners encounter more complex and nuanced realities in professional HCI work across diverse sectors and problem spaces. Problem areas in specific sectors such as health-care, education, and sustainability now demand a more holistic, cross-disciplinary understanding of people's experiences and design within the broader ecosystems. These challenges involve diverse, interconnected stakeholders, various types of knowledge, and the integration of multiple tools, materials, and spaces. HCI education needs to reflect the complex and cross-disciplinary dynamics of HCI, exploring ways to contribute meaningfully to converging problem spaces across different landscapes [8, 12].

More immersive and experiential pedagogies have been explored [22, 32, 40]. For instance, with the recognized discrepancy between HCI learning and the expectations of the UX industry, HCI pedagogy has integrated innovative learning techniques ranging from workshops, studio-based learning to industrial internships to enhance important UX skills [16]. Also, to better reflect the emergent, reconstructive, and collaborative nature of HCI and design pedagogy [19], other studies have adopted adding participatory [2] and user-engaged research [3, 33]. However, current pedagogical approaches struggle to connect HCI practice and research with broader real-world complexities, inadequately conveying the implications of this interconnectedness for HCI practice. A more holistic view of the HCI ecosystem across diverse problem spaces helps students reflect on how human practices and values manifest, preparing mindsets to tackle the multifaceted challenges in professional HCI work.

To this end, this study is guided by the following research questions:

- What pedagogical strategies effectively bridge the gap between HCI pedagogy and the complex, interconnected nature of real-world HCI and design challenges?
- How do these strategies influence students' learning experiences, facilitating students to understand HCI and design principles and processes within broader ecosystems of a chosen problem space?

3 Methods

We examined this phenomenon in a semester-long HCI and design course offered in a large research university in North America. Based on our field study, which utilizes data from weekly in-class fieldnotes, instructor interviews, and observations of teacher planning and debrief sessions, we first identified three pedagogical features that enhance student learning in HCI and design. Then,

we inductively identified three separate themes related to student experiences through a thematic analysis of end-of-semester student interviews and biweekly reflections. These findings were compared and interpreted in relation to one another. Our evidence-based study provides practical suggestions on how the implemented pedagogical strategies have facilitated specific learning experiences for students.

3.1 Context of the study

The course was offered by an innovative school (we call it "the School") within the university, a school that integrates technology, design, and business of innovation. It adopted a Challenge-Based Reflective Learning (CBRL) framework [37], where students explore real-world challenges through reflective practice across domains (in this case, technology, design, and business). CBRL engages students with diverse challenges using a context-driven, experiential learning approach. This framework shifts the focus from content-oriented learning to an iterative process in which students practice and achieve cross-domain mastery. It equips students with domain-general competencies (applicable and transferable across various domains, such as abstract computation and design strategy) and domain-specific skills (specific to particular fields). The CBRL framework emphasizes the importance of context in shaping the relevance of problems, fostering students to understand sociotechnical realities by transcending disciplinary boundaries [37].

13 students (we call them "students" instead of "participants" of the study) in this 16-week course were guided by three main course instructors (we call them "instructors" instead of "participants" of the study to distinguish the two different types of our participants and their roles) with design, research, and materials science expertise. Students in this course worked with a local community center (we call it "the Center") serving underrepresented youth (we call them "youth," "users," or "end-users" interchangeably) in the community. The Center featured a physical technology space that offered diverse educational programs in computing, design and making, and entrepreneurship for underserved youth. Students were tasked to ideate and design the Center spaces (i.e., maker space and materials science library) and create an 18-week after-school curriculum utilizing these spaces (e.g., for digital fabrication, product design, materials science applications). The details of their design tasks are discussed in the pedagogy section of the findings.

3.2 Participants

13 students participated in this field study. Of these, 10 students were interviewed at the end of the semester to share their experiences and insights. Among the students interviewed, seven were undergraduates and three were graduate students at the university. All but one student were pursuing integrated degree programs offered by the School within the university.

Based on student reports on their academic or professional background and interests, the students demonstrated various product and digital fabrication skills, including user interface/user experience (UI/UX) design (5 out of 10), 3D modeling (3 out of 10), virtual reality (VR) (3 out of 10), and product management and marketing skills (5 out of 10). They also reported different levels of proficiency

that involved the chosen challenge space: product and spatial design, materials science, and learning sciences.

3.3 Data Sources and Analysis

For this article, we used student and instructor interviews, observational fieldnotes taken during class sessions (i.e., 2-hours weekly) and other instructor meetings, and archives (i.e., instructional materials, such as slides, syllabus, and other instructional materials) for analysis. First, we analyzed 10 student interviews conducted at the end of the semester. Each semi-structured interview lasted approximately 30 to 45 minutes, and they were audio recorded and transcribed. The authors engaged in a separate open coding process to identify initial patterns using the terms used by the participants [36]. We then used thematic analysis [4, 15] to identify 2nd-level themes together during multiple rounds of peer debriefing meetings. The final codebook contained five themes (i.e., physical/material, relational, reflective, learning, teamwork) and their distinct codes (e.g., “materials, tools, and machines,” “figuring out from ambiguity,” “product design or design thinking/ process,” and “meeting multiple/diverse stakeholders’ needs”).

Then, the first author, who took weekly fieldnotes in the classroom with another observer, used these codes to make sense of the themes and extract instances from the 166 pages of the fieldnotes. These findings from the fieldnotes complement the interview data [24], utilizing heuristics and depicting a balanced account of instructors’ strategies and students’ practice-of-knowing [26, 29] in this experiential learning environment.

4 Findings

To address our two research questions, we structure our findings into two sections, highlighting three pedagogical strategies and corresponding student learning experiences. The three emerged themes related to pedagogical strategies are: 1) embedding HCI within a cross-disciplinary problem space, 2) engaging stakeholders beyond end-users, and 3) utilizing cross-functional team structure for collaboration. Since these strategies are within the broader CBRL framework focusing on challenge-driven, practice-oriented learning [37], it is also appropriate to understand them in relation to one another as a broader pedagogical approach.

4.1 Pedagogical Strategies

4.1.1 Embedding HCI within a cross-disciplinary problem space. A distinctive pedagogical feature of this course was embedding the HCI and design problem and related learning activities within a broader cross-disciplinary problem space. The initial problem space was mutually defined by the Center stakeholders and the instructors based on the Center’s needs and the student’s learning goals and expertise: to “develop and deploy a product innovation 18-week curriculum for high school students utilizing the integrated functionality of the Lab.”

Through a series of discussions and planning sessions, this problem yielded multiple interrelated deliverables and expectations, especially as the instructors planned ways to embed students’ design activities while identifying other cross-disciplinary learning (e.g., material science and learning sciences) into the problem space. Thus, while the final project goal was to ideate and design a maker

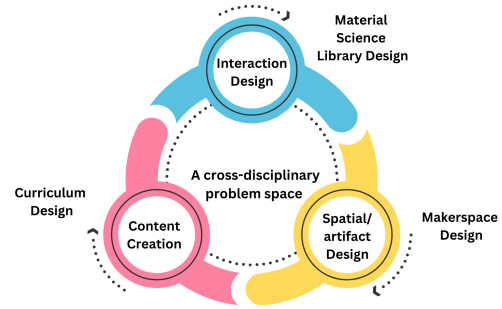


Figure 1: Embedding HCI and design within a broader cross-disciplinary problem space.

space and materials science library for youth, students needed to understand the content that the End-Users would learn in the space (i.e., materials science and product design), plan the learning activities the Users would engage in while interacting with machines and materials (e.g., laser cutters, 3D printers, sewing machines), and identify needed tools and techniques that Users will interact in these spaces (e.g., software for the machines like Adobe illustrator for vector sketching). Within this problem space, students needed to grasp relevant technology, design, and materials science concepts, such as foundational material properties, criteria for material selections, types of machines for digital fabrication, and science learning standards for high school students in the region.

The instructors incorporated a range of HCI and design methods such as user research, prototyping, and spatial and interactive design. They also ensured that students seek specific knowledge related to materials science, product design, and learning sciences. For instance, students explored questions from, “What kind of machines and tools are needed for high school learning in the maker space and the materials library?” “What categories of materials should be in the library collection?” “How do you categorize and store them?” to “What are considered safe tools for youth?” To help students address a diverse range of cross-disciplinary questions, others were present in the classroom including a maker-in-residence from the School (maker space education and design expert) and the Center stakeholders (curriculum design and education experts, those with knowledge of the physical spaces). Figure 1 represents the chosen cross-disciplinary problem space where students’ practice of HCI and design were explored.

4.1.2 Engaging Stakeholders Beyond End-Users. In this course, the stakeholders’ close interaction with students in the classroom played a pivotal role in shaping the design process. Stakeholders and relevant communities of practice exist on many different levels beyond a single group—in organizations, broader infrastructure, and society [9, 14]. Departing from traditional client-based classroom projects where client interactions are typically limited to the beginning and the end of the semester, the instructors integrated several stakeholders from the Center throughout the entire course, inviting a program officer, senior director, curriculum designer and instructor, and other Center advisors. One or two stakeholders were present in the classroom throughout the entire semester, helping students

explore the problem and solution from various angles. As many served as teachers and educators themselves, they were motivated to help students learn through the process, not to simply complete the projects.

During Week 2, students visited the site to learn about the space and interact with staff. Students conducted user study with the Center youth in Week 5. During week 14, students interacted with the parents of the youth participating in the Center programs. In Week 11 and for the final presentation, students visited the site again to present and receive feedback on their design and curriculum recommendations from the stakeholders and sponsoring industry partners.

Engaging stakeholders beyond end-users in this course served two functions. First, it was to help students acquire contextual and practical knowledge involving the immediate HCI and design work (e.g., safety and operation measures, spatial constraints). Second, it was designed to aid students in gaining domain-specific knowledge, which impacted how they make design decisions. For instance, students worked with stakeholders to think through how material properties can be learned by high school youth by utilizing digital fabrication tools in the maker space (e.g., CNC machine, laser cutting machine). Because the presence of such knowledge was not immediately apparent to students, the stakeholders helped through questioning in design ideation and prototyping.

4.1.3 Utilizing Cross-Functional Team Structure for Collaboration. Finally, the instructors implemented a cross-functional team structure in the classroom to facilitate students' cross-disciplinary problem exploration. Cross-functional teams, commonly used in organizations (e.g., agile projects, software development, interdisciplinary innovation), represent different knowledge domains and work areas to accomplish a joint task [21]. 13 students were divided into three functional teams (i.e., the maker space team (n=5), the library team (n=4), and the curriculum team (n=4)). Given the small class size, the instructors initially debated whether to implement this strategy but decided to use it, aiming to expose students to focused, in-depth work within their functional teams as well as the broader, interconnected work across all three teams. Thus, each of the functional team objectives were the following:

- The makerspace team - designing the maker space for the Center, responsible for the overall spatial design, selection of machines and tools used in the space
- The library team – designing the material science library for the Center, responsible for the overall spatial design of the library space, designing a storage and retrieval system for the materials collections
- The curriculum team – designing an integrated curriculum for youth, focusing on product design and material science; responsible for utilizing the functionality of the two spaces and the embedded tools and machines to support youth's integrated learning

During Week 3, students shared in class their completed “background sheet,” expressing their strengths and interests along with relevant skills and knowledge pertaining to each functional team. The instructors wrote these on a whiteboard, placed each student's name in one or two functional teams. Students then adjusted and finalized their team assignments over the following two weeks.

4.2 Student Learning Experiences

Our findings revealed that the three pedagogical strategies offer significant benefits for student learning in the HCI and design classrooms. First, students reported gaining domain-specific knowledge that extend beyond HCI principles and the design process. Additionally, this pedagogical approach fostered a deeper understanding of end-user needs while helping students recognize the interests of multiple stakeholders beyond the end-users, who also shape the problem space and design decisions. Many students emphasized that working with real-world stakeholders provided invaluable preparation for their future careers. Finally, students highlighted the challenges, ambiguity, and uncertainty they faced while working cross-functionally on a cross-disciplinary challenge, as they sought a way to effectively make sense of the challenge together and collaborate across functional boundaries.

4.2.1 Grasping domain-specific knowledge in the broader challenge space. In this course, students shared that they not only learned HCI principles and design processes but also gained domain-specific knowledge involving the complex cross-disciplinary challenge. Rather than through direct instruction of these disciplinary concepts, students learned through their embodied interaction and engagement within the problem space. As one student, T1-S3-UG noted,

“I learned a lot about material science without knowing that I was technically learning about material science. When I'm building things, I like seeing different options for, like, sticking two objects together to build a strong structure or something, making joints or using glue, or doing both at the same time what different types of glue are there? What different types of paints are there for different materials, and why does it matter?” (T1-S3-UG)

This finding highlights how embedding HCI within a cross-disciplinary problem space allows students to acquire domain-specific knowledge organically through hands-on engagement. As demonstrated by T1-S3-UG's reflection, students grasped disciplinary concepts such as material science through their practical interactions with design challenges, deepening their understanding of how a HCI and design problem might lend itself to specific disciplinary content.

4.2.2 Recognizing co-evolving dynamics of HCI and design through stakeholder engagements. Students' main takeaways from this theme included gaining a deeper understanding of end-users' needs, recognizing the dynamics of diverse stakeholders, and navigating real-world project workflows. Students considered the broader dynamics of the problem space by seeking information from the involved stakeholders including the Center staff, parents, and the users. This approach allowed their design recommendations to extend beyond a single audience, recognizing the complexities involving stakeholders constituting the given ecosystem. As T2-S6-G mentioned: *“It's been interesting to navigate the balance between the different stakeholders—students, the “corporate suits,” and those evaluating me as a professor. Each group has their own needs. Students want one thing, but then you have to consider the people in charge, too. It becomes an optimization problem: who do you prioritize? Do you aim to keep students happy, ensure the center's staff are satisfied, or focus on the instructor who's grading you? It's a bit of a conundrum figuring out who to really listen to.”*

T1-S2-UG learned to apply some of the stakeholders' expectations into the iterative design processes, such as incorporating their day-to-day operation of the space, safety constraints, and regulatory requirements. T1-S2-UG said: *"So one thing is like, you know, a lot of these machines that are cutting things, require a lot of ventilation. On top of that, we want it to be closed. such as 3D printers. A lot of these, like printers such as Prusa, they're open and for places where, you know, young children could be. We wanted them to be closed. So we chose machines that were like fully enclosed CNC machines that were like those on top of that."*

At the same time, students recognized, and were challenged by, high ambiguity in this process, as stakeholders had open expectations about the space and curriculum design. As T3-S8-G noted about "progressing forward" with little "blueprint" or "specifications" from the stakeholders, the student said: *"...that forward progress, although it may be confusing, it might be stressful at some time, ... just forces you to create and... establish rules and... guidelines that aren't necessarily there in a traditional setting."* This realization was a critical learning experience for many students as they learned that the stakeholders' decisions depended on the quality and the direction of their ideas, and vice versa. This theme highlights how the scope of the problem space expands and evolves based on the dynamics of the involved stakeholders, opening and changing the scope of the design beyond predetermined and single-user-focused design activities.

Students gained additional knowledge from various stakeholders, enhancing their understanding of the iterative, complex, and collectively constructive design process. Other key learning included: regional science learning standards (from an educational advisor), multipurpose functionality of the space (from the Center director), and practical teaching considerations (from a program manager) (i.e., deciding between 18-week curriculum vs. shorter week modules).

In sum, iterating with diverse stakeholders exposed students to the evolving, and sometimes conflicting dynamics of the HCI and design process. The majority of students noted that this experience provided preparation for future professional careers where understanding and integrating diverse expectations are essential for successful project outcomes.

4.2.3 Aligning processes and work across the functional teams. Students recognized the necessity of collaborating across functions in this problem space, particularly in the latter half of the semester. Prior to this transition, they primarily worked within assigned teams, participating in class-wide discussions to understand the design challenge while producing work for their functional teams.

Post their official presentation to the stakeholders during Week 11, students realized the importance of holistic alignment of their work across teams. While many thought their own functional teams "worked well," figuring out each person's roles, who is in charge, and how to coordinate. However, this not apparent for their cross-functional work. T1-S2-UG observed: *"Every single group has our own sort of process and our own sort of different workflows. And it's, in the beginning, everything was just like, every group focuses on their own thing. But when we merge, we realized, oh, oh, there's a lot of differences between our groups. And I think going over the hurdle was a really good experience."* Some felt that this process was like

"chaos" (T3-S10-UG), not knowing how each function with distinct needs and knowledge areas should align with one another to create a coherent product.

To ease this process, students strategized by choosing a specific functional work domain to anchor and create a process for other domains. Thus, content creation (i.e., curriculum and learning design, See Figure 1) became central to determining the spatial and interaction design within the physical spaces. T2-S6-G described this understanding: *"So, we are now just focusing more on what curriculum the team is creating, what materials are needed, and what tools are needed. The storage space capacity for that. And like where to buy these? Materials. I think the Lab created a really great spreadsheet of materials that they need for the room. And so we're kind of using that."*

Overall, students gained domain-specific knowledge related to the broader context in which HCI is embedded, recognized the complexity that various stakeholders bring to the design ecosystem, and learned to collaborate and interconnect knowledge and skillsets across different domains of work. As T3-S9-UG summed up the learning in this course:

"I think especially with like this job economy and stuff, they just want you to do way more than what it used to. It's not just like interviewing anymore. So I think that [this course is] really cool and will be so beneficial to students. I would say another takeaway for me is... being able to coordinate between a curriculum team and then actually building out a material science library and the [makerspace] lab for it all to work in tandem... just seeing the dynamics of that and making sure it's not all too separated."

The interplay of diverse disciplinary knowledge, stakeholders, and functional roles impacted their learning, as they made sense of different perspectives, areas of knowledge, and interests of the groups involved.

5 Discussion

Our findings demonstrate the value of embedding HCI and design within real-world, cross-disciplinary challenges. This approach expands HCI education beyond isolated activities centered on single users or context, exposing students to the multifaceted nature of real-world design and implementation [39].

We offer practical recommendations for designing a course that integrate HCI into cross-disciplinary problem spaces. First, embedding HCI within broader challenges requires instructors to move beyond focusing solely on HCI principles as learning goals. Part of the learning goals should also focus on how students can *apply* the learned principles to different ecosystems and sectors such as healthcare, education, and real estate. Challenge-based learning [13], along with its broader pragmatic and constructive educational approaches [10, 31, 35], including the CBRL framework [37], is increasingly recognized as essential in HCI and design education, prompting suggestions to incorporate diverse pedagogies and design methods [11, 19, 41]. While experimenting with such approaches may challenge the ongoing discussion on what topics and methods are considered core to HCI [8], it helps students apply HCI to various problems, better preparing them for the changing dynamics of professional HCI and design work [14, 16, 18]. For these pedagogical approaches, instructors must become proficient

in facilitating learning in unfamiliar areas, serving as facilitators to help students discern, seek, and integrate the necessary knowledge across various disciplines and sectors. In the context of transdisciplinary education [37], co-teaching and co-mentoring have been proposed and shown to be beneficial [20]. Collaborative course design practices, and fostering a culture of collaboration among instructors with diverse domain expertise, as frequently exhibited in this case, are instrumental in bridging HCI principles and practices with other sectors and addressing complex challenge areas.

Second, to ensure a productive partnership between community or industry stakeholders and the university, instructors should outline and monitor expectations early on while allowing flexibility and space for iterative learning. A large part of the learning objectives should focus on students engaging with contextual knowledge and learning to navigate competing interests that often shape HCI practice. As ongoing stakeholder engagement can be resource-intensive, alternative methods from sociotechnical research and social informatics (e.g., SCOT–Social Construction of Technology, Pinch & Bijker [30]) [1, 6] can be complemented to prepare students for real-world complex projects. These frameworks can help students to move away from quick technological fixes to exploring more sustainable and ethical solutions.

Finally, to utilize cross-functional team structures in the classroom and reflect the interdisciplinary nature of real-world HCI practice, instructors should provide scaffolding throughout the course to help students define roles, establish communication norms, and coordinate across teams. Cross-functional collaboration is known to be difficult, as team members must navigate ambiguous and interrelated structures while overcoming boundaries and barriers stemming from different languages, practices, and specialty areas—differences rooted in distinct thought worlds [5, 23]. As demonstrated in this case, it is critical to normalize ambiguity and uncertainty in cross-functional work. Instructors should explicitly outline, from the beginning, what it means to work in a cross-functional team, including the potential experiences and outcomes, such as tensions, misalignments, and strategies to address them. Making these dynamics explicit allows students to process their experiences in a structured and reflective manner. Instructors should also provide an initial framework for establishing communication norms. These strategies include allocating time for students to learn about each other's interests and strengths when forming teams and, and over time, supporting teams in renegotiating goals and developing clear expectations about who *knows* and *acts* what [21]. These interventions help students engage with the problem space while learning to develop collective structures, such as a transactive memory system (TMS)—a shared memory system for coordinating cross-functional work [21]—to effectively manage the complexity and uncertainty inherent in cross-functional collaboration.

6 Implication and Future work

Current HCI pedagogies often fall short of reflecting the collaborative and complex nature of HCI practice, especially given the rapid changes in innovation structures facing the future of work. This calls for the HCI community to explore how HCI education can reflect and address these changes. Classroom constraints – such as confined learning objectives, timelines, and institutional norms –

often conflict with these realities of professional practice, despite the call to align HCI theory and practice. Further work is needed to integrate diverse stakeholder groups and disciplinary knowledge areas, providing authentic learning experiences that equip students with holistic approaches for the evolving HCI work.

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